

VADITA: Visual Analytics for Detecting Illegal Transport Activities

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Figure 1: The user interface consists of the following components: (A) Filter Panel, (B) Legend Panel, (C) Cluster View, (D) Map View (d1 for aggregation, d2 for focus), (E) Timeline View (e1 shows the transport data, e2 shows the aggregated transport data and focus, e3 shows the harbor data, e4 shows the association between vessels and commodities, e5 shows the transaction data), and (F) Statistic View (f1 shows the average length of stay in a region, f2 shows the aggregated OD (Origin-Destination) data, f3 shows the focused OD data).

ABSTRACT

This paper presents a visual analytics system designed to address the IEEE VAST Challenge 2024 Mini-Challenge 2. The system can support the matching and anomaly detection of multi-source heterogeneous spatio-temporal data, thereby enabling the detection of illegal transport activities. The primary contribution of the system lies in its analysis-driven interaction design.

Index Terms: spatio-temporal data, anomaly detection, visual analytics.

1 INTRODUCTION

The IEEE VAST Challenge 2024 Mini-Challenge 2¹ requires participants to develop visual analytics tools to analyze spatio-temporal data in CatchNet. In more detail, first associate vessels with their probable commodities and then detect illegal transport activities through four questions (Q1-4). The provided data is a composite of graphs from various sources, representing a temporal dataset concerning vessels and commodities. The nodes in the dataset include entities such as vessels, locations, commodities, and delivery reports, while the edges encompass events such as transactions, vessel movements, and harbor reports. We interpret this

challenge as a spatio-temporal data visual analytics task [1, 2]. The challenge can be divided into two parts: the Q1 can be understood as a matching problem involving multi-source heterogeneous spatio-temporal data, while the Q2-4 can be viewed as anomaly detection problems involving the same data type. For the first part, we utilized visual analytics methods to comprehend the data, uncover associations, and ultimately design a matching algorithm to accomplish the Q1. For the second part, we developed a visual analytics system to assess vessel behavior from multiple dimensions and identify anomalies.

2 SYSTEM DESIGN

Our system interface (Fig. 1) is mainly built with D3.js [3] and Vue.js. The interface consists of several components, including the Filter Panel (Fig. 1.A), Legend Panel (Fig. 1.B), Cluster View (Fig. 1.C), Map View (Fig. 1.D), Timeline View (Fig. 1.E), and Statistic View (Fig. 1.F).

Filter Panel. The Filter Panel allows users to filter by vessel, location, commodity, and time interval attributes. Specifically, we allow users to filter for vessels and locations based on types.

Legend Panel The Legend Panel centrally displays the color coding for the attributes of vessels, locations, and commodities, thereby reducing visual redundancy in other views.

Cluster View. In the Cluster View, each point represents a vessel, with color indicating the vessel type. Vessels closer together on the plot are likely to have more similar behavior patterns. Users can select a specific vessel by clicking on it to focus, or they can brush to select a group of vessels for further analysis. We implemented

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¹<https://vast-challenge.github.io/2024/MC2.html>

the Cluster View by first filtering and aggregating vessel movement data based on specific date ranges, vessels, and locations. We then constructed a time series for each vessel, capturing daily movement counts and dwell times at various locations. To visualize these patterns, we calculated the Dynamic Time Warping (DTW) distance matrix between the time series and applied t-SNE for dimensionality reduction. This process enabled us to project the complex, high-dimensional time series data into a 2D space, facilitating the visualization of vessel clusters based on their movement behaviors.

Map View. The Map View displays geographic information and the frequent movement paths of vessels. Users can hover their mouse over different locations to view detailed information about those locations and the vessel movement paths. To reduce visual clutter, we filter out less frequent movement paths. In this view, d1 represents aggregated information about the vessels selected by the user either through brushing in the Cluster View or through the Filter Panel, while d2 provides detailed information about a specific vessel that the user has focused on in the Cluster View.

Timeline View. The Timeline View visualizes multi-source data over time, using a stacked layout to enable quick access to multidimensional information about vessels and commodities within specific intervals, facilitating the rapid identification of anomalous behaviors. The x-axis represents time across all subviews, while the y-axis represents vessels in e1–e4 and commodity types in e5. Brushing is available on all axes to help users quickly focus on specific vessels, commodities, and time periods.

e1 displays transport data for selected vessels, with rectangle length indicating dwell time and color representing location. e2 includes two time strips: the upper strip aggregates transport data for selected vessels, showing the most common location at any time, while the lower strip shows data for the focused vessel. This helps users identify vessel movement patterns and detect anomalies by comparing the focused vessel's behavior against typical patterns. e3 shows harbor data for all selected vessels, encoded similarly to e1. e4 visualizes the association between vessels and commodities; rectangles represent commodities, and dashed lines between them indicate the possible period during which the commodities were acquired, with the color of the dashed lines representing the location. e5 presents aggregated transaction data, with color intensity reflecting the number of transactions on a given day. This setup enables comprehensive temporal analysis, supporting the detection of abnormal vessel activities.

Statistic View. The Statistic View presents various statistical insights on vessel dwell times at different locations and OD behavior.

f1 uses stacked bar charts to show the average dwell time of vessels in likely fishing regions. The left bar, with diagonal stripes, represents aggregated data for all selected vessels, while the right bar represents the focused vessel. f2 and f3 use chord diagrams to visualize OD behavior and dwell time proportions at different locations, with f2 showing aggregated data for all selected vessels and f3 for the focused vessel. These insights help identify potential anomalous fishing activities based on vessel movement and dwell patterns.

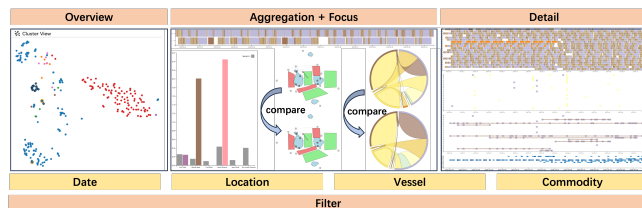


Figure 2: System workflow. Users filter across different dimensions to explore from an overview to details.

3 WORKFLOW

Our system enables users to filter across different data dimensions, facilitating an exploration process that moves from an overview

to aggregation and focus, and finally to detailed analysis (Fig. 2). Based on this workflow, our system supports multiple methods for identifying vessels with potential anomalous behavior. In the Overview step, users can brush or click to select significant clusters or outliers. In the Aggregation+Focus step, users can analyze these selections to identify individuals within the clusters that differ significantly from the aggregated pattern or to determine the reasons behind the outlier status. Finally, in the Detail step, users can confirm their hypotheses using more granular information. Users can also apply filters on different attributes at any step to observe resulting changes, enabling a deeper understanding of the data.

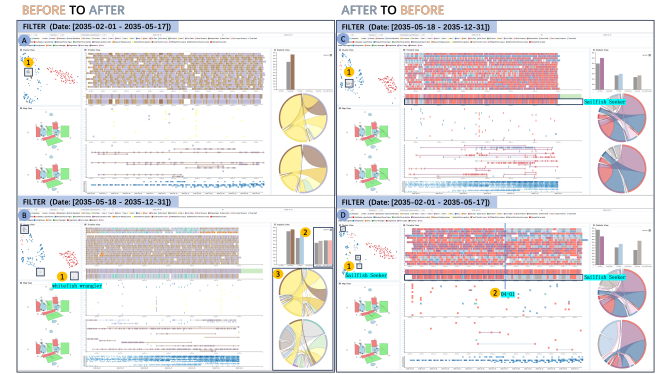


Figure 3: Comparison of vessels' fishing activity behavior during different periods before (A, D) and after (B, C) the capture.

4 ILLUSTRATIVE CASE

In the question 4, we aim to identify suspicious behavioral changes in vessels following the capture of SouthSeafood Express Corp. We illustrate the system's capability through a specific case study.

In Fig. 3.A, we selected a relatively isolated cluster (Fig. 3.A1). Upon adjusting the timeframe to the after-capture period, the system revealed the scenario shown in Fig. 3.B. Notably, the vessel Whitefish, previously central to the cluster, now appears as an outlier (Fig. 3.B1). A detailed analysis across multiple dimensions indicated that while Whitefish's average dwell time in Ecological Preserves and other behavioral metrics remained consistent with normal vessels (Fig. 3.B2), its dwell time distribution exhibited a significant increase in visits to Nemo Reef (Fig. 3.B3). This suggests that after capture, vessels involved in illegal fishing reduced their dwell time in Ecological Preserves to align with legitimate vessels but compensated by increasing the frequency of their visits.

In Fig. 3.C, we filtered a cluster representing normal behavior (Fig. 3.C1) and examined the before-capture period. As shown in Fig. 3.D1, the cluster split into two sub-clusters. The behavior in the upper left sub-cluster remained typical, while the other sub-cluster, including vessels like Sailfish Seeker, displayed anomalous behavior with prolonged dwell times at Nemo Reef, primarily after April 1st (Fig. 3.D2). This pattern indicates that these vessels engaged in illegal activities prior to the capture but reverted to normal fishing activities upon becoming aware of the captures.

We therefore conclude that a new suspicious fishing behavior involves reducing dwell time in Ecological Preserves while simultaneously increasing the frequency of visits to these areas.

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